NeuroData Quality AI

Explainable Neuroimaging Dataset Quality Control and Research-Readiness Prediction Platform

Complete Project Details and System Design

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Objective: Build an AI/ML-first platform that evaluates neuroimaging datasets for quality, consistency, anomalies, metadata completeness, and research readiness before they are used for machine-learning research.

1. Executive Summary

NeuroData Quality AI is a research-support platform that identifies poor-quality, inconsistent, incomplete, biased, or anomalous neuroimaging data before model training. It does not diagnose patients and does not reproduce brain mapping or clinical decision-support products. The core predictions are produced by independently developed PyTorch and Scikit-learn models. Google Gemini is used only to explain structured results and generate readable research reports.

2. Problem Statement

Neuroimaging datasets may contain motion artifacts, blur, noise, corrupted files, inconsistent resolutions, missing metadata, duplicate scans, scanner differences, class imbalance, and unusual outliers. These issues can reduce model performance, introduce bias, and make research difficult to reproduce.

3. Project Goals

Develop a deep-learning scan-quality classifier; detect artifacts; identify anomalies; analyze metadata; measure class and site imbalance; produce a research-readiness score; explain predictions using Grad-CAM and feature importance; integrate Gemini for reporting; and deploy the system with Flask.

4. Main System Output

The user uploads MRI/fMRI files and optional metadata. The system returns scan-level quality labels, confidence scores, artifact indicators, anomaly flags, metadata warnings, dataset statistics, a readiness score, and recommended cleaning actions.

5. AI/ML Module 1: Image-Quality Classification

Use PyTorch with transfer learning such as ResNet, EfficientNet, or MobileNet. Classify scans as good, acceptable, poor, or unusable. Apply resizing, normalization, intensity standardization, and augmentation. Evaluate using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.

6. AI/ML Module 2: Artifact and Motion Detection

Detect blur, noise, motion-related distortion, abnormal intensity, cropping, missing slices, and inconsistent dimensions using OpenCV, NumPy, image statistics, and a CNN-based classifier.

7. AI/ML Module 3: Anomaly Detection

Use Isolation Forest, One-Class SVM, autoencoder reconstruction error, PCA, or embedding distance to identify scans that differ from the majority of the dataset.

8. AI/ML Module 4: Metadata and Bias Analysis

Check missing values, invalid fields, image dimensions, voxel spacing, acquisition parameters, scanner information, duplicate records, train/validation/test separation, class distribution, age distribution, sex categories, sites, and scanner types.

9. AI/ML Module 5: Research-Readiness Model

Combine quality indicators such as poor-scan percentage, motion score, artifact probability, metadata completeness, outlier percentage, class imbalance, and scanner inconsistency. Use Random Forest, Logistic Regression, or Gradient Boosting to classify a dataset as Ready, Requires Cleaning, or Not Ready.

10. Explainable AI

Use Grad-CAM to highlight image regions influencing quality predictions. Use feature importance or SHAP-style analysis to show why the dataset received its readiness score. The system must separate model findings from recommendations.

11. Gemini API Role

Gemini receives structured prediction JSON and generates explanations, research-readiness summaries, technical definitions, cleaning recommendations, and answers about the analysis. Gemini must not make the core quality prediction.

12. Gemini Data Flow

MRI files → preprocessing → trained ML models → structured JSON → Gemini explanation → report displayed in the website. Never expose the API key in frontend JavaScript.

13. Website Pages

Home; Dataset Upload; Scan Quality; Dataset Analysis; Research Readiness; AI Research Assistant; Model Evaluation; Prediction History; and Report Download.

14. System Architecture

User → HTML/CSS/JavaScript interface → Flask backend → file validation → preprocessing → PyTorch quality model + Scikit-learn anomaly model + metadata analyzer → feature fusion → readiness model → explainability layer → Gemini report generator → website/PDF report.

15. Technology Stack

Python; PyTorch; Scikit-learn; Pandas; NumPy; OpenCV; Matplotlib; Flask; HTML5; CSS; JavaScript; SQLite or SQL; Google Gemini API through Google AI Studio; Git; GitHub; and Hugging Face Spaces.

16. Suggested Folder Structure

neurodata-quality-ai/; app.py; requirements.txt; .env; routes/; ml/; explainability/; services/; models/; datasets/; templates/; static/; database/; notebooks/.

17. Example API Endpoints

POST /api/upload — validate dataset. POST /api/analyze — run analysis. GET /api/scan/ — retrieve scan results. GET /api/dataset/ — retrieve dataset results. POST /api/explain — generate Gemini explanation. GET /api/report/ — generate report. GET /api/history — list previous analyses.

18. Gemini Setup

Install google-genai and python-dotenv. Store GEMINI_API_KEY in .env. Load it from the environment in Flask. Keep .env in .gitignore. Use a stable Gemini model available to the account, such as Gemini 2.5 Flash, for concise explanations and reports.

19. Example Gemini Code

from google import genai; import os; from dotenv import load_dotenv; load_dotenv(); client = genai.Client(api_key=os.getenv('GEMINI_API_KEY')); response = client.models.generate_content(model='gemini-2.5-flash', contents=prompt); return response.text

20. Dataset Strategy

Begin with a public, appropriately licensed neuroimaging dataset. If quality labels are unavailable, create controlled training examples by applying documented blur, noise, motion, intensity, and missing-slice distortions to clean images. Start with 2D slices and expand to 3D volumes only after the first pipeline works.

21. Security and Privacy

Use anonymized or public data. Do not upload patient-identifiable information in the demo. Validate file size and type. Do not send unnecessary identifiers to Gemini. Keep the API key server-side. Display a research-only disclaimer.

22. Real-World Usefulness

The system can pre-screen datasets before expensive training, identify scans for manual review, support multi-site comparison, improve reproducibility, reduce corrupted training data, and provide a consistent quality-control checklist for research teams.

23. Difference from BrainSightAI

This is not a brain-mapping, diagnostic, or patient-specific clinical-analysis platform. It focuses on dataset quality control, anomaly detection, metadata consistency, bias checks, and ML research readiness. It is complementary to neuroimaging research pipelines rather than a copy of the company's core activities.

24. Development Roadmap

Phase 1: project setup and upload pipeline. Phase 2: train image-quality model. Phase 3: add artifact and anomaly detection. Phase 4: add metadata and imbalance analysis. Phase 5: train readiness model. Phase 6: add Grad-CAM and feature importance. Phase 7: integrate Gemini. Phase 8: add reports, testing, GitHub documentation, and deployment.

25. Demo Scenario

Upload a small anonymized dataset containing clean and artificially degraded scans. The platform classifies quality, detects outliers, checks metadata, calculates readiness, displays explanations, and generates a Gemini-written research report.

26. Resume Value

The project demonstrates Python, PyTorch, Scikit-learn, computer vision, deep learning, data preprocessing, feature engineering, anomaly detection, predictive modeling, model evaluation, Flask, SQL, Gemini API integration, GitHub, and deployment.

27. Limitations

This is a research-support prototype, not a clinical diagnostic device. Quality scores depend on labels and datasets. Synthetic distortions may not represent every real artifact. Anomaly flags require human review, and bias analysis identifies potential imbalance rather than proving clinical fairness.

28. Final Project Definition

NeuroData Quality AI is an explainable multimodal AI/ML platform that evaluates neuroimaging datasets for image quality, artifacts, anomalies, metadata completeness, class imbalance, and research readiness. PyTorch and Scikit-learn perform the core predictions; Flask serves the models; and Gemini explains structured results and generates reports.